

ANSHUMAAN ANAND

Bhubaneswar, India | +91 8630600086 | anshumaananand2005@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

KIIT University, Bhubaneswar, India

B.Tech in Computer Science & Engineering | **CGPA: 8.10**

Expected 2027

Delhi Public School, Varanasi, India

Grade 12 (CBSE): 80.6% | Grade 10 (CBSE): 86.2%

2019 – 2022

SKILLS

Languages: C, Java, Python, SQL, JavaScript

Full-Stack Web: HTML, CSS, React 19, FastAPI, Django, REST APIs, MongoDB, MySQL, PostgreSQL

Data Analytics & ML: Python, Pandas, NumPy, scikit-learn, TensorFlow, Keras, Matplotlib, Statistical Modeling

Geospatial: MapLibre GL JS, Leaflet.js, PostGIS, GeoJSON, Turf.js, GeoTIFF, Spatial Data Processing

BI & Visualization: Power BI, Dashboard Development, Business Intelligence, Data Visualization

Tools & Platforms: Git, Docker, Socket.IO, Zustand, Vite, LangGraph, Redis

Concepts: Data Structures, Algorithms, OOP, DBMS, System Design, WebSocket / Real-Time Systems, Agent Architecture

EXPERIENCE

Research Intern – IIT Kharagpur

May 2026 – Jun 2026

Seismic Hazard Simulation & AI-Enhanced Rapid Nowcasting Project

- Built interactive geospatial maps for real-time seismic hazard visualization using MapLibre GL JS and Leaflet.js, rendering dynamic hazard layers, intensity contours, and sensor overlays on live data feeds.
- Designed and managed PostGIS/PostgreSQL spatial databases to ingest and query earthquake, soil classification, and IoT sensor data at scale; authored spatial SQL for proximity, buffering, and raster intersections.
- Contributed to a Multi-Task AI/ML framework fusing smartphone accelerometers, fiber-optic (DAS) networks, and SAR satellite imagery into unified early-warning disaster insights.
- Developed nowcasting pipelines that convert real-time seismic signals into actionable emergency resource allocation recommendations, cutting analyst response time for field teams.

PROJECTS

AI Agent with Memory & Tool Use

[In Progress]

Python, FastAPI, LangGraph, Groq (Llama 3), PostgreSQL, React 19, Vite

- Building a conversational AI agent with persistent memory across sessions using PostgreSQL and LangGraph state graphs, enabling context-aware multi-turn dialogue with per-user memory isolation.
- Implementing a tool-calling router with four agent tools: DuckDuckGo web search, URL summarization, sandboxed Python code execution, and vector-based memory retrieval — agent selects tools dynamically based on query intent.
- Designing a FastAPI backend with async endpoints and React + Vite frontend; deploying on Render and Vercel using Groq-hosted Llama 3 as the free LLM backbone with streaming response support.

Hazard Map – Real-Time Disaster Response GIS Dashboard

React 19, Vite, FastAPI, MapLibre GL JS, PostgreSQL/PostGIS, WebSocket, Docker

- Engineered a dual-mode disaster simulation dashboard: offline mode runs GMPE-based earthquake and landslide models entirely in-browser; live mode polls USGS WebSocket feeds and broadcasts spatial events over a job queue with background workers.
- Built interactive hazard visualizations (hazard rings, heatmaps, aftershock clusters) rendered on a WebGL MapLibre canvas, with click-to-simulate epicenter selection and real-time layer updates.
- Implemented a FastAPI backend with PostgreSQL/PostGIS for spatial queries, a Redis-backed job queue for async simulation tasks, and a trained ML model for landslide susceptibility scoring.
- Containerized the full stack (React + FastAPI + PostgreSQL) using Docker Compose with separate service networking and environment-based config.

Attendance Risk Prediction System

Python, Pandas, scikit-learn, FastAPI

- Engineered an end-to-end ML pipeline for student attendance risk classification — feature engineering from behavioral and academic data, model training with scikit-learn, and real-time inference via FastAPI REST endpoints.
- Evaluated Logistic Regression, Random Forest, and Gradient Boosting models; selected best performer using precision, recall, and F1 scores, achieving reliable identification of at-risk students before attendance drops critically.
- Performed thorough data preprocessing with Pandas: handled missing values, encoded categorical features, and normalized inputs to ensure clean, consistent training data across all model runs.

ACHIEVEMENTS & CERTIFICATIONS

- Power BI Professional Training — 60-hour certified course by ExcelR in association with KIIT University (Feb–Mar 2026); Certificate ID: 56093/ExcelR.
- Solved 250+ LeetCode problems (DSA); completed Web Development Bootcamp (Udemy); currently pursuing Data Science & ML coursework covering Python, Pandas, scikit-learn, and predictive modeling.
- Participated in Amazon Hackathon (2026), Kasparov Hackathon, and Smart India Hackathon (SIH) internal round — competitive programming and rapid product-building under time constraints.